



Alessandro Chessari

Postdoctoral Researcher in Computational Quantum Physics

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Education

Commissariat à l'Énergie atomique (CEA) / Université Grenoble Alpes <i>PhD in Theoretical Physics</i>	<i>Grenoble, France</i> 2022 – 2025
École Normale Supérieure de Lyon <i>MSc in Theoretical Physics</i> ◦ M1 with High Honors	<i>Lyon, France</i> 2020 – 2022
Politecnico di Milano <i>MSc in Physics Engineering</i> ◦ Final grade: 110/110	<i>Milano, Italy</i> 2019 – 2022
Politecnico di Milano <i>BSc in Physics Engineering</i>	<i>Milano, Italy</i> 2016 – 2019

Academic Positions

Postdoctoral Researcher <i>TU Wien</i> ◦ Quantum-inspired tensor network algorithms. ◦ Advisors: Dr. Anna Kauch, Dr. Markus Wallerberger.	<i>Vienna, Austria</i> Dec 2025 – present
Visiting PhD Student <i>TU Wien</i> ◦ Quantics Tensor Trains for one-dimensional spin chains.	<i>Vienna, Austria</i> Dec 2024 – June 2025
PhD student <i>Commissariat à l'Énergie atomique (CEA)</i> ◦ PhD Thesis: <i>From single-qubit to many-body readout in circuit QED platforms.</i> ◦ Advisors: Dr. Michele Filippone, Dr. Yann-Michel Niquet.	<i>Grenoble, France</i> Oct 2022 – Oct 2025
Research Intern <i>Commissariat à l'Énergie atomique (CEA)</i> ◦ Master Thesis: admittance of a dissipative quantum many-body system. Supervisor Dr. Michele Filippone.	<i>Grenoble, France</i> Mar 2022 – July 2022
Research Intern <i>CRMN, ENS de Lyon</i> ◦ M1 Thesis: polarisation transfer between nuclear spins in paramagnetic solids (NMR). Supervisor Dr. Quentin Stern.	<i>Lyon, France</i> May 2021 – July 2021

Research Intern

SemiSpin Lab, Politecnico di Milano

Milano, Italy

June 2019 – July 2019

- Bachelor Thesis: optically generated spin currents in Ge detected via the inverse spin-Hall effect in Pt/Cu/Ge and Au/Cu/Ge. Supervisor Prof. Federico Bottegoni.

Fellowships and Awards

- **CEA PACE Award** (2024–2025): research visit to TU Wien, funded by Commissariat à l'Énergie atomique (CEA).
- **IDEX Formation Scholarship** (2024–2025): research visit to TU Wien, funded by Université Grenoble Alpes.
- **Double Degree Program** (2020–2022): Politecnico di Milano – ENS de Lyon, Erasmus Scholarship.
- **IdexLyon Excellence Scholarship** (2020–2021): excellence program funded by ENS de Lyon and Université Claude Bernard Lyon 1.

Publications

Efficient Optimization of Tensor Rings with Low-Rank Environments

M. Jeannin, **A. Chessari**, J. von Delft

arXiv:2610.10027 (2026)

Many-Body Admittance Spectroscopy

A. Chessari, M. Filippone

In preparation (2026)

From single qubit to many-body readout in Circuit Quantum Electrodynamics platforms

A. Chessari

PhD Thesis, Université Grenoble Alpes (2025)

Parametric longitudinal coupling of a semiconductor charge qubit and an rf resonator

V. Champain, S. Zihlmann, **A. Chessari**, B. Bertrand, H. Niebojewski, E. Dumur, X. Jehl, V. Schmitt, B. Brun, C. Winkelmann, Y.-M. Niquet, M. Filippone, S. De Franceschi, R. Maurand

Phys. Rev. Applied 23, 034067 (2025)

Unifying Floquet theory of longitudinal and dispersive readout

A. Chessari, E. A. Rodríguez-Mena, J. C. Abadillo-Uriel, V. Champain, S. Zihlmann, R. Maurand, Y.-M. Niquet, M. Filippone

Phys. Rev. Lett. 134, 037003 (2025)

Role of electron polarization in nuclear spin diffusion

A. Chessari, S. F. Cousin, S. Jannin, Q. Stern

Phys. Rev. B 107, 224429 (2023)

Talks

QUAST Workshop on Tensor Compression

Talk: *Bethe Ansatz for the Heisenberg XXX model*

Vienna, Austria

24 Jul 2026

Internal seminar at TU Wien

Talk: *DMRG: from matrix product state to the finite-size algorithm*

Vienna, Austria

22 Jul 2026

QTech-BudapestAccepted talk: *Unifying Floquet theory of longitudinal and dispersive readout*

Budapest, Hungary

24 Sep 2025

Internal seminar at TU WienTalk: *Unifying Floquet theory of longitudinal and dispersive readout*

Vienna, Austria

31 Mar 2025

Participation in Schools and Conferences

QUAST Workshop on Tensor Compression

TU Wien

Vienna, Austria

23–24 Jul 2026

Workshop on the Interplay between Neural and Tensor Networks

Maison du Quantique Alpes, Université Grenoble Alpes

Grenoble, France

6–10 Jul 2026

Workshop on Tensor Cross Interpolation and Learning Tensor Networks

Maison du Quantique Alpes, Université Grenoble Alpes

Grenoble, France

6–10 Oct 2025

QTech-Budapest

Hungarian Academy of Sciences

Budapest, Hungary

22–25 Sep 2025

Silicon Quantum Electronics Workshop (SiQEW)

Davos, Switzerland

4–6 Sep 2024

Trimester: Quantum Many-Body Systems Out of EquilibriumThematic trimester program at *Institut Henri Poincaré (IHP)*.

Paris, France

8 Jan – 22 Mar 2024

Introductory School: Quantum Many-Body Systems Out of EquilibriumIntroductory school of the IHP thematic trimester at *Institut d'Études Scientifiques de Cargèse (IESC)*.

Cargèse, France

26 Nov – 2 Dec 2023

GDR Quantum Mesoscopic PhysicsConference on mesoscopic physics at *CNRS Centre Paul Langevin*.

Aussois, France

28 Nov – 1 Dec 2022

GDR IQFA 13Conference on quantum engineering, from fundamental aspects to applications, at *Centre de Nanosciences et de Nanotechnologies (C2N)*.

Saclay, France

16–18 Nov 2022

Doctoral Training: Frontiers of Condensed MatterSelected participant of the doctoral training school at *École de Physique des Houches*.

Les Houches, France

10–21 Oct 2022

Winter School on Ultracold Quantum GasesSelected participant. School organized by Prof. Francesca Ferlaino, Prof. Rudolf Grimm and Prof. Hanns-Christoph Nägerl at the *University of Innsbruck*.

Innsbruck, Austria

9–12 Feb 2022

Hyp21 International Hyperpolarization ConferenceInternational conference on hyperpolarization methods in NMR, EPR and MRI, organized by *Université Claude Bernard Lyon 1*.

Lyon, France

5–9 Sep 2021

Teaching

Exercise sessions, Electrical Phenomena and Transport

Université Grenoble Alpes

Sep–Dec 2024

Exercise sessions, Electrical Phenomena and Transport

Université Grenoble Alpes

Sep–Dec 2023

Skills

Programming: Julia, Python, C, C++, MATLAB, LaTeX

Languages: Italian (native), Romanian (native), English (fluent), French (fluent), German (B2)